

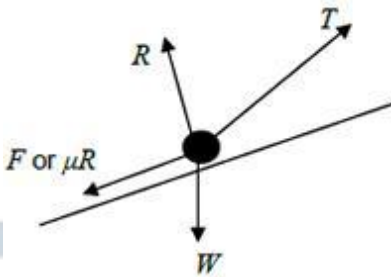
Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses model for maximum friction = μmg	AO3.3	B1	$F_{\max} = \mu mg$ $= 0.85 \times 20 \times 9.8$ $= 166.6 \text{ N}$
	Makes an appropriate comparison	AO1.1a	M1	
	Explains clearly why crate remains stationary	AO2.4	E1	$150 < 166.6$ $\therefore$ crate does not move
(b)	Forms an equation by resolving vertically Condone one of sign error or cos error	AO3.1b	M1	$20g = R + 150\sin 15^\circ$ $R = 157.177 \text{ N}$
	Obtains correct reaction force	AO1.1b	A1	$F_{\max} = \mu \times 157.177$ $= 133.6 \text{ N}$
	Uses maximum friction = μR With 'their' reaction force Must identify maximum or limiting friction	AO1.1b	B1F	$150\cos 15^\circ = 145 \text{ N}$ $145 > 133.6$ $\therefore$ crate begins to move
	Compares $150\cos 15^\circ$ with 'their' maximum friction	AO1.1a	M1	
	Explains, using their values, why the crate begins to move.	AO2.4	E1F	
				Total 8 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Resolves horizontally and vertically to obtain expressions for F and R (allow consistent mixing of sin and cos)	AO3.4	M1	Resolving vertically $R = 8 \times 9.8 - 50\sin 40^\circ$ Resolving horizontally $F = 50\cos 40^\circ$
	Obtains correct expressions for R and F	AO1.1b	A1	

	States friction model $F = \mu R$ with 'their' values for F and R	AO1.2	B1	$F = \mu R$ $50 \cos 40^\circ = \mu(8 \times 9.8 - 50 \sin 40^\circ)$ $\mu = \frac{50 \cos 40^\circ}{8 \times 9.8 - 50 \sin 40^\circ}$
	Completes a rigorous argument that results with correct μ AG Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips	AO2.1	R1	$\mu = 0.8279660445 = 0.83$ (2sf) AG
(b) (i)	Draws correct diagram with exactly four forces showing arrow heads and labels Can use Mg or $8g$ or 78.4 for W	AO3.3	B1	
(ii)	Resolves perpendicular to the plane resulting in a three term equation containing; R , $8g \cos 5^\circ$ and $T \sin 40^\circ$ (or $T \cos 50^\circ$) OR Resolves horizontally and vertically to obtain equations of motion in horizontal and vertical directions	AO3.1b	M1	$R = 8 \times 9.8 \cos 5^\circ - T \sin 40^\circ$ $= 78.1 - T \sin 40^\circ$
	Obtains correct expression for R OR Obtains correct horizontal and vertical equations	AO1.1b	A1	$T \cos 40^\circ - 8 \times 9.8 \sin 5^\circ - F = 8 \times 3$ $T \cos 40^\circ - 8 \times 9.8 \sin 5^\circ - 0.827 \dots \times (8 \times 9.8 \cos 5^\circ - T \sin 40^\circ) = 24$
	Forms a four term equation of motion parallel to the plane with correct terms (allow sign errors) OR Eliminates R to solve for T	AO3.1b	M1	$T = \frac{24 + 8 \times 9.8 \sin 5^\circ + 0.827 \dots \times 8 \times 9.8 \cos 5^\circ}{\cos 40^\circ + 0.827 \dots \times \sin 40^\circ}$ $T = 73.55939193 = 74$ (2sf)
	Obtains correct equation of	AO1.1b	A1	

motion OR Obtains correct expression(s) without R		
Uses the friction model in the four term equation of motion where R is in the form $a - bT$ and a and b are positive constants OR Uses the friction model in the horizontal and vertical equations Dependent on previous M1	AO3.1b	dM1
Solves for T	AO1.1b	A1
Obtains correct value of T with 2 sf accuracy. FT incorrect value found for T provided both M1 marks and dM1 mark have been awarded	AO3.2a	A1F
Total 12 marks		

Q3.

(a)

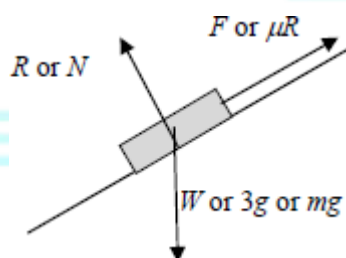


Diagram with exactly three forces showing arrow heads and labelled. If components are also shown they must use a different style e.g. dashed lines then they can be ignored. Friction must be up the slope.

B1

1

(b) $(R =) 3 \times 9.8 \cos 40^\circ = 22.5 \text{ N}$

M1: Resolving perpendicular to the slope. Can use $\sin 40^\circ$ or $\cos 50^\circ$ for method mark, with g or 9.8 .

A1: Correct normal reaction. Accept
AWRT 22.5 (Note use of 9.81 still gives 22.5 N.)

M1A1

2

(c) $(F =) 0.2R = 4.50 \text{ N}$

M1: Use of $F = \mu R$.

A1F: Correct friction. Accept 4.5 N or AWRT 4.50.

(Accept 4.51 N from the use of 9.81.)

Follow through incorrect normal reaction from part (b).

M1A1F

2

(d) $3a = 3 \times 9.8 \sin 40^\circ - 4.504$

M1: Three term equation of motion with correct terms, with $3a$, either component of weight and their answer to part (c) for F .

A1F: Equation of motion with correct terms and signs.

M1A1F

$a = 4.80 \text{ m s}^{-2}$

A1F: Correct acceleration. Accept 4.8 or AWRT 4.80.

(Note that using 9.81 still gives 4.80 m s^{-2}).

Follow through friction from part (c).

A1F

3

- (e) No air resistance force acting
or
No other forces acting on the box.
or
They (forces in the diagram) are the only forces that act.

OR

No turning effect (due to forces).

or

Forces are concurrent. OE

B1: Correct assumption.

Ignore irrelevant comments

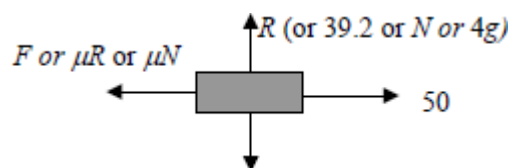
B1

1

[9]

Q4.

(a)



mg or W or $4g$ or 39.2

B1: Correct force diagram with four forces with arrows

*and labels. Accept words eg friction instead of letters.
 Ignore negative signs in labels.
 Do not accept 4 kg for the weight.
 Award marks if forces are drawn on the diagram in the question.*

B1
1

(b) 39.2 N

*B1: Correct reaction force.
 Accept 4g. Do not accept 39.*

B1
1

(c) $50 - F = 4 \times 3$

*M1: Three term equation of motion with the correct terms.
 A1: Correct equation with correct signs.*

M1A1

$$F = 38$$

A1: Correct friction.

A1
3

(d) $38 = \mu \times 39.2$

M1: Use of $F = \mu R$ with their answers to (b) and (c).

M1

$$\mu = \frac{38}{39.2} = 0.969$$

*A1F: Correct μ based on their answers to (b) and (c).
 Accept AWRT 0.969.*

A1F

*Note: $F = 12$ leads to 0.306 and award M1 A1F
 Condone 0.97 or FT to 2sf
 Condone use of inequalities.*

2

(e) Less friction, so a smaller coefficient of friction.

B1: Less friction.

B1

B1: Smaller μ .

B1

Note:

More friction anywhere scores B0 B0

Less friction, greater μ scores B1 B0

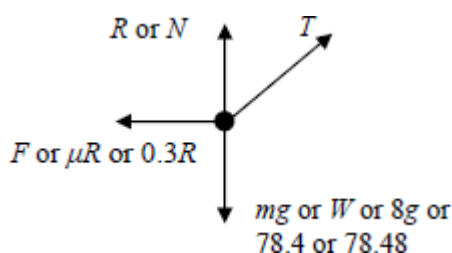
Smaller μ with no / inexact reason B0 B1

2

[9]

Q5.

(a)



B1: Diagram with **exactly four** forces showing arrow heads and labelled. If components are also shown and they use a different style, eg dashed lines, they can be ignored.

Note: Award mark if forces drawn on the diagram in the question.

Note: Do not accept 8 kg for the weight.

Note Accept μR or $0.3R$ for F .

B1

1

(b) $R + 40 \sin \theta = 8 \times 9.8$

M1: Resolving vertically to obtain a three term equation, with R , $T \sin$ or $\cos(30^\circ$ or $60^\circ)$ and $8g$ oe.

A1: Correct equation

M1A1

$$R = 78.4 - 40 \sin \theta$$

A1: Correct expression for R .

Accept $(R=)8g - T \sin 30^\circ$

Note if using $g = 9.81$ accept

$$R = 78.48 - 0.5T \text{ or } R = 78.5 - 0.5T$$

A1

3

(c) $8a = 40 \cos \theta - 0.3(78.4 - 40 \sin \theta)$

M1: Use of the friction inequality with R from part (b)

A1: Correct equation.

M1A1

$$8a = -23.52 + 40 \cos \theta + 12 \sin \theta$$

M1: Use of equation of motion with $40 \sin \theta$ or $40 \cos \theta$ and $8a$ and a friction term.

M1

$$p = -2.94$$

$$q = 8$$

$$r = 1.5$$

A1: One correct value.

A1

A1: All three values correct.

A1

5

[9]

Q6.

(a) $18g - T = 18a$

M1: Three term equation of motion for the 18 kg particle.

A1: Correct equation of motion for the 18 kg particle.

(Accept $T - 18g = 18a$)

M1A1

$$T = 12a$$

B1: Equation of motion for the block that has signs consistent with the first equation.

B1

$$18g - 12a = 18a$$

$$a = \frac{18g}{30} = 5.88 \text{ ms}^{-2}$$

*A1: Correct acceleration from correct work. Accept $\frac{3g}{5}$
Do not penalise consistent use of negative acceleration, provided final answer positive.*

Special Case:

*Whole String Method $18g = 30a$ and $a = \frac{18g}{30} = 5.88$
OE M1A1*

Note using $g = 9.81$ gives 5.89, also accept 5.88.

A1

4

(b) (i) $18g - T = 18 \times 3$

M1: Three term equation of motion for the 18 kg particle with $a = 3$ seen.

A1: Correct equation.

M1A1

$$T = 18g - 18 \times 3 = 122(.4)\text{N}$$

A1: Correct tension. Accept 122.4.

Note using $g = 9.81$ gives 123, also accept 122.

A1

3

(ii) ($R = 12 \times 9.8 = 117.6 \text{ N} \Rightarrow 118 \text{ N (to 3sf)}$)

B1: Correct normal reaction. Accept 117 and 117.6. Final answer must be positive.

Do not accept 12g.

Note using $g = 9.81$ gives 118, also accept 117.

B1

1

(iii) $122.4 - F = 12 \times 3$

M1: Three term equation of motion for the block, containing their tension, F and 12×3 .

A1F: Correct equation. Follow through T from part (b)(i).

M1A1F

$$F = 86.4$$

A1F: Candidate's T minus 36.

A1F

$$86.4 = \mu \times 117.6$$

dM1: Use of $F = \mu R$ with AWRT 117 or 118 for R and the candidate's value of F provided positive.

dM1

$$\mu = \frac{86.4}{117.6} = 0.735$$

A1: Correct μ . Accept anything between 0.728 and 0.739 inclusive. Allow 0.73 and 0.74.

A1

5

- (c) No air resistance or no other forces
B1: One assumption from list

B1

Horizontal String

Block is a particle or **they** are particles

B1: For another assumption from list.

Do not penalise assumptions not in the list.

B1

2

[15]

Q7.

- (a) $T = 2 \times 9.8 = 19.6 \text{ N}$

M1: Equating tension and weight.

A1: Correct tension CAO

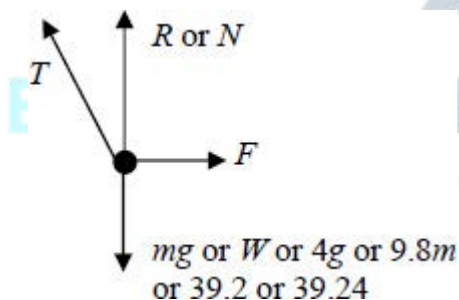
Accept $2g$

Accept 19.62 from $g = 9.81$

M1A1

2

- (b)



B1: R , F (not μR) and mg correct

B1

B1: T correct, must be in roughly correct direction.

If more than four forces shown, do not award more than one mark.

Note all forces must be shown as arrows and have labels.

Note some candidates may draw the force diagram in the section with the question.

Components can be ignored if shown in a different notation eg dashed arrows.

B1

2

(c) $T \cos 30^\circ + R = 4 \times 9.8$

*M1: Resolving vertically to form a three term equation.
(May be implied)*

M1

$(R =) 39.2 - 19.6 \cos 30^\circ$
 $= 39.2 - 16.9741\dots$

*A1 Correct expression for R or equation for R.
Must see 19.6cos30 or equivalent (eg 2gsin60)*

A1

$= 22.2259\dots$

$= 22.2 \text{ N (to 3sf)}$ AG

A1: Correct force. Must see intermediate working, for example third or fourth line of working in solution opposite.

Example: $19.6 \sin 30^\circ - R = 4 \times 9.8$ scores M1A0A0.

Use of $g = 9.81$ still gives 22.2 N as the final answer.

A1

3

(d) $T \cos 60^\circ = F$

M1: Resolving horizontally

M1

$F = 19.6 \cos 60^\circ = 9.8$

A1: Correct expression for friction

A1

$19.6 \cos 60^\circ \leq \mu(39.2 - 19.6 \cos 30^\circ)$

dM1: Use of $F = \mu R$ or $F \leq \mu R$ (do not allow $F \geq \mu R$)

dM1

$$\mu \geq \frac{19.6 \cos 60^\circ}{39.2 - 19.6 \cos 30^\circ}$$

$\mu \geq 0.441$

A1: Final answer of $\mu = 0.441$ or $\mu \geq 0.441$ from correct working

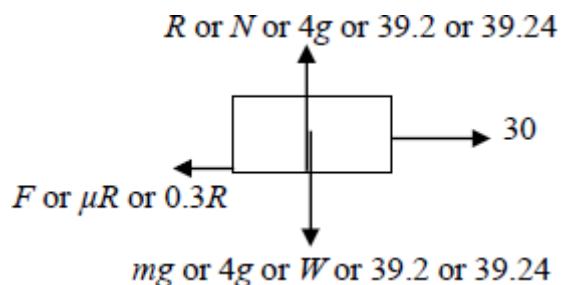
Use of $g = 9.81$ still gives 0.441 as the final answer.

A1

4

Q8.

(a)



B1: Diagram with four forces showing arrow heads and labelled.

Ignore negative signs in labels.

Note: Award mark if forces drawn on the diagram in the question.

Note: Do not accept 4 kg for the weight.

Note Accept μR for F .

B1

1

(b) $(R = 4 \times 9.8 =) 39.2 \text{ N}$

B1: Correct normal reaction. Accept $4g$

B1

1

(c) $(F =) 0.3 \times 39.2 = 11.76 = 11.8 \text{ N (to 3sf)}$

M1: Use of $(F =) \mu R$

M1

A1: Correct friction.

Accept $1.2g$ or 11.7 or 11.76 N .

Do not condone further work after the value for friction has been obtained.

A1

2

(d) $4a = 30 - 11.76$

M1: Three term equation of motion.

A1F: Correct equation.

M1A1F

$$a = \frac{30 - 11.76}{4} = 4.56 \text{ ms}^{-2}$$

A1F: Correct acceleration.
FT candidates F from part (c).
Accept 4.55 from 11.8.

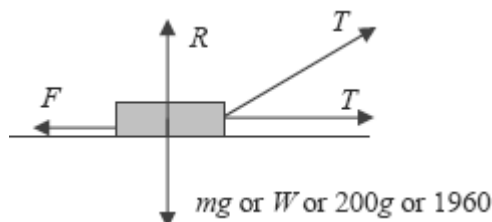
A1F

3

[7]

Q9.

(a)



B1: F , R and mg (or equivalent) with arrows and labels.

*B1: Two **equal** tension forces with arrows and labels.*

Ignore components if shown with a different notation, eg dotted lines.

B1

B1

2

(b) $R + T \sin 20^\circ = 1960$

OR

$R + T \sin 20^\circ = 200g$

M1: Resolving vertically with three terms.
Must include $\sin 20^\circ$ or $\cos 20^\circ$ or $\sin 70^\circ$ or $\cos 70^\circ$ with T and $200g$ or 1960 .

A1: Correct equation.

M1A1

$(R =) 1960 - T \sin 20^\circ$

OR

$(R =) 200g - T \sin 20^\circ$

A1: Correct expression for the normal reaction.

Note: If $g = 9.81$ is used for the first time deduct one mark. Should get 1962 instead of 1960

A1

3

(c) $T \cos 20^\circ + T - F = 200 \times 0.3$

M1: Four term equation of motion. Must include $\sin 20^\circ$ or $\cos 20^\circ$ or $\sin 70^\circ$ or $\cos 70^\circ$ with T and a second T term with no trig.
A1: Correct equation

M1A1

$$T \cos 20^\circ + T - 0.4 (1960 - T \sin 20^\circ) = 200 \times 0.3$$

M1: Use of friction law with their expression for R , provided that R has two terms. Note that this mark does not depend on any previous marks.

Example:

If Candidate gives 1960 as answer to part (b), then: $F = 0.4 \times 1960 = 784$ scores M0 here

M1

$$T = \frac{60 + 784}{\cos 20^\circ + 1 + 0.4 \sin 20^\circ} = 406$$

dM1: Solving for T .

dM1

Note: This mark requires both of the previous M marks.

A1: Correct tension.

Accept AWWF 406 to 407.

Note: If $g = 9.81$ is used should get 407 instead of 406.

A1

5

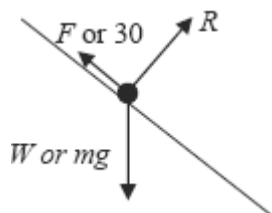
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[10]

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Q10.

(a)



B1: Correct force diagram with arrows and labels.

B1

1

(b) $30 = mg \sin 42^\circ$

M1: Two term equation by resolving parallel to the slope. Must include g .

A1: Correct equation.

M1A1

$$m = \frac{30}{9.8 \sin 42^\circ} = 4.57$$

A1: Correct value for m .

A1

3

(c) $(R =) mg \cos 42^\circ = 33.3 \text{ N}$

M1: Resolving perpendicular to the slope to find R . Must include g .

A1: Correct value.

M1A1F

2

(d) $30 = \mu R$

$$\mu = \frac{30}{R} = 0.900$$

M1: Use of $F = \mu R$

M1

M1: Solving for μ .

M1

A1F: Correct value for μ . Accept 0.9.

A1F

3

Use of $g = 9.81$

(b) 4.57 B1

(c) 33.3 B1

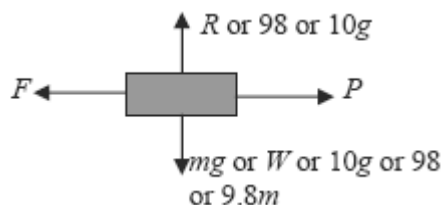
(d) 0.900 B1

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[9]

Q11.

(a)



B1: Correct force diagram with arrows and labels.

Note: Award mark if forces drawn on the diagram in the question.

Note: Do not accept 10 kg for the weight.

Note: Do not accept μR or $0.5R$ for F .

B1
1

(b) (i) $(R = 10 \times 9.8 =) 98 \text{ N}$

*B1: Correct normal reaction. Accept $10g$.
No need to see the letter R or working.*

B1
1

(ii) $(F \leq) 0.5 \times 98$

$(F \leq) 49$

*B1: Correct maximum value for friction.
Accept $5g$.
No need to see the letter F or any
working. Ignore any inequalities.
For FT, must be 0.5 of candidate's answer to (b)(i).*

B1F
1

(iii) $(F =) 30 \text{ N}$

B1: Correct friction. Allow -30 .

B1
1

(c) $80 - 49 = 10a$

*M1: Three term equation motion,
containing 80, candidate's 49 and $10a$
(not $10ga$) in any combination.
A1F: Correct equation including signs.*

M1A1F

$a = 3.1 \text{ m s}^{-2}$

*A1F: Correct acceleration.
FT candidate's answer to (b)(ii).*

A1F
3

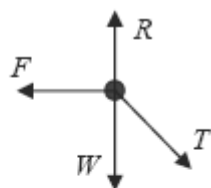
Allow use of $g = 9.81$

(b) (i) 98.1	B1
(b) (ii) 49.05 or 49.1 or 49	B1
(c) 3.095 or 3.09 or 3.1	M1A1A1

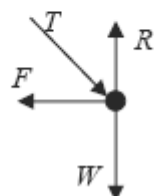
[7]

Q12.

(a)



OR



B1: Diagram with four forces showing arrow heads and labelled.

Allow mg or $8g$.

Allow T or 40 or other reasonable notation.

Allow μR .

Direction of friction must be to the left.

Any components must be shown in a different style.

B1

1

(b) $8g + 40 \sin 30^\circ (= R)$

M1: Expression for normal reaction, with mg or $8g$ and $40 \sin 30^\circ$ or $40 \cos 30^\circ$.

Allow incorrect signs.

M1

A1: Correct expression with correct signs.

A1

$(R =) 98.4 \text{ N}$ **AG**

A1: Correct value from correct working.

Use of $g = 9.81$ gives 98.5 N . Do not penalise if you have already done so earlier in the scripts.

Otherwise penalise by 1 mark.

A1

3

(c) $F = 40 \cos 30^\circ = 34.6 \text{ N}$

M1: Use of $40 \cos 30^\circ$ or $40 \sin 30^\circ$.

Award M0 if any extra terms

M1

A1: Correct value for friction. Don't need to see F .

A1

(d) $40 \cos 30^\circ \leq \mu \times 98.4$

M1: Use or $F \leq \mu R$ (or $F = \mu R$). Must use $R = 98.4$ and a positive value for F .

M1

*A1F: Correct inequality or equation
Allow use of $F = \mu R$ throughout.*

A1F

$$\mu \geq \frac{40 \cos 30^\circ}{98.4}$$

$$\mu \geq 0.352$$

A1F: Correct minimum value. For follow through must use $R = 98.4$ and their value for F from part (c). For example use of $\sin 30^\circ$ in part (c) gives 0.203.

A1F

3

[9]

Q13.

(a) (i) $20 \times 9.8 = R + 60 \sin 30^\circ$

M1: Equation or expression for normal reaction with mg or $20g$ or 196 and $60 \sin 30^\circ$ or $60 \cos 30^\circ$.

A1: Correct equation or expression with correct signs.

M1A1

$(R =) 20 \times 9.8 - 60 \sin 30^\circ = 166 \text{ N}$ AG

*A1: Correct value from correct working.
Must be positive.*

Don't penalise use of $g = 9.81$ if already done earlier on script. Should still get 166, but from 166.2.

A1

3

(ii) $166\mu = 60 \cos 30^\circ$

*M1: Use of $F = \mu R$, with $R = 166$ or 166.2 . Do not allow inequalities here.
M1: Resolving horizontally with $\cos 30^\circ$*

or $\sin 30^\circ$ oe

A1: Correct equation

Examples:

$$166\mu = 60 \text{ M1M0A0}$$

$$166\mu = -60 \cos 30^\circ \text{ M1M1A0}$$

M1M1A1

$$\mu = \frac{60 \cos 30^\circ}{166}$$

$$= 0.313$$

A1: Correct coefficient of friction.

A1

4

(b) $20 \times 0.8 = T \cos 30^\circ - 0.313(20 \times 9.8 - T \sin 30^\circ)$

B1: $20g - T \sin 30^\circ$ oe seen.

M1: Three term equation of motion,
where normal reaction is dependent on T .

A1F: Correct equation

B1M1A1F

$$T = \frac{20 \times 0.8 + 0.313 \times 20 \times 9.8}{\cos 30^\circ + 0.313 \sin 30^\circ} = 75.6 \text{ N}$$

dM1: Solving for T including factorisation.

A1F: Correct tension.

AWRT 75.6

Follow through incorrect values of μ
from part (a).

Don't penalise use of $g = 9.81$ if already
done earlier on script. Should get 75.7.
Allow 75.8 if intermediate values rounded.

5

[12]

Q14.

(a) $R = 15 \times 9.8 = 147 \text{ N}$

$$F = 0.2 \times 147 = 29.4 \text{ N}$$

Finding normal reaction.

M1

Use of $F = \mu R$

M1

Correct friction force.

A1

3

(b) $5g - T = 5a$

Three term equation of motion for 5 kg particle

M1

Correct equation

A1

$T - 29.4 = 15a$

AG

Three term equation of motion for 15 kg block

M1

$49 - 29.4 = 20a$

Correct equation

A1

$a = 0.98 \text{ m s}^{-2}$

Correct final answer from correct working.

A1

5

(c) $1.4^2 = 0^2 + 2 \times 0.98s$

$s = \frac{1.4^2}{1.96} = 1 \text{ m}$

Use of a constant acceleration equation with $u = 0$

M1

Correct equation

A1

Correct distance

A1

3

[11]

Q15.

(a) (i) $0.2a = -0.2 \times 9.8 \sin 20^\circ$

AG

$a = -9.8 \sin 20^\circ = -3.35 \text{ m s}^{-2}$

Two term equation of motion with weight

resolved

M1

Correct equation

A1

Correct acceleration from correct working

A1

*SC No negative sign but otherwise correct
award M1A1A0*

Allow $a = -g \sin 20^\circ$

3

(ii) $0 = 4^2 + 2 \times (-3.35)s$

*Use of constant acceleration equation
with $v = 0$ and $u = 4$*

M1

Correct equation

A1

$$s = \frac{16}{6.7} = 2.39 \text{ m}$$

Correct distance

A1

3

- (iii) The puck slides back down the slope as the puck is at rest and the resultant force is now acting down the slope / no friction / smooth slope.

Slides back down

B1

Acceptable explanation

E1

2

(b) (i) $R = 0.2 \times 9.8 \cos 20^\circ$ **AG**

*Finding normal reaction by resolving.
Must see a trig term.*

M1

$$F = 0.5 \times 0.2 \times 9.8 \cos 20^\circ$$

Use of $F = \mu R$

M1

$$= 0.921 \text{ N}$$

Correct friction from correct working.

A1

3

(ii) $0.2a = -0.921 - 0.2 \times 9.8 \sin 20^\circ$

Three term equation of motion with the weight resolved

M1

Correct equation

A1

$$a = -7.96 \text{ m s}^{-2}$$

Correct acceleration (with or without the minus sign, applied to both A1 marks)

A1

3

- (iii) The puck stays at rest because the friction has a maximum of 0.921 and the component of the weight down the slope is less (0.670)

Stays at rest

B1

Acceptable explanation

dE1

2

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[16]

Q16.

(a) $F = 0.4 \times 1000 \times 9.8$

Use of $F = \mu R$

M1

$$= 3920$$

AG

Correct friction from correct working.

Allow $F = 0.4 \times 9800$

Allow verification

A1

2

(b) $P - 3920 = 5000 \times 0.8$

Three term equation of motion including an explicit 0.8

M1

Correct equation

A1

$$P = 7920 \text{ N}$$

AG

*Correct force from correct working.
Allow $P = 5000 \times 0.8 + 3920$*

A1

3

(c) $T - 3920 = 1000 \times 0.8$

Three term equation of motion

M1

Correct equation

A1

$$T = 4720 \text{ N}$$

Correct tension

A1

or

$$7920 - T = 4000 \times 0.8$$

$$T = 4720 \text{ N}$$

3

- (d) Friction is reduced because the normal reaction is reduced.

Friction reduced

B1

Acceptable explanation

E1

2

[10]

Q17.

(a) $2 \times 9.8 - T = 2 \times 0.9$

Three term equation of motion for B

M1

Correct equation

A1

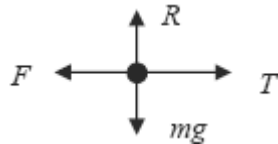
$$T = 19.6 - 1.8 = 17.8 \text{ N}$$

Correct tension

A1

3

(b)



Correct force diagram with labels and arrows

B1

1

(c) $R = 3 \times 9.8 = 29.4 \text{ N}$

Calculation of normal reaction force.

B1

1

(d) $17.8 - F = 3 \times 0.9$

Three term equation of motion for A.

M1

$$F = 17.8 - 2.7 = 15.1 \text{ N}$$

Correct friction

EXAM PAPERS PRACTICE

A1

2

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(e) $15.1 = 29.4\mu$

Use of $F = \mu R$

M1

$$\mu = \frac{15.1}{29.4} = 0.514$$

Correct value for μ

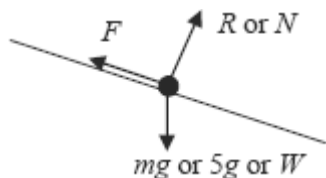
A1

2

[9]

Q18.

(a)



Correct force diagram with labels and arrows
 Accept components of the weight if
 shown in a different notation with the
 weight also shown.
 B0 if components are shown instead of
 the weight.

B1

1

(b) $(R =) 5 \times 9.8 \cos 40^\circ = 37.5 \text{ N}$ **AG**

Attempt at resolving perpendicular to the
 slope (eg $49 \sin 40^\circ$)

M1

Correct value from correct working

A1

2

(c) $5 \times 0.8 = 5 \times 9.8 \sin 40^\circ - \mu \times 5 \times 9.8 \cos 40^\circ$

Use of $F = \mu R$ at any stage and with any
 F but with $R = 37.5$ OE

B1

Three term equation of motion from
 resolving parallel to the slope with weight
 component, friction and ma term.

M1

Correct terms seen (may be as 31.5,
 37.5μ (or F) and 4)

A1

Correct signs

A1

$$\mu = \frac{5 \times 9.8 \sin 40^\circ - 5 \times 0.8}{5 \times 9.8 \cos 40^\circ} = 0.733$$

Solving for μ

m1

A1: Correct value for μ

$\frac{11}{15}$
 Allow 0.732 but not $\frac{11}{15}$ unless converted to a decimal

A1

6

- (d) There is less friction so the coefficient of friction must be less.

Less friction

B1

*Smaller coefficient of friction
If the answer and explanation contradict each other, award no marks*

B1

2

[11]

Q19.

- (a) The string is light and inextensible or inelastic or taut

First assumption

B1

Second assumption

B1

2

- (b) $6 = 0 + 4a$

Finding a using a CA equation

M1

$$a = \frac{6}{4} = 1.5$$

Correct a from correct working

A1

2

- (c) $7 \times 9.8 - T = 7 \times 1.5$

Three term equation of motion with F for the 7 kg particle. Correct equation

M1A1

$$T = 68.6 - 10.5 = 58.1$$

Correct tension

A1

3

(d) $58.1 - F = 13 \times 1.5$

Three term equation of motion with F for the 13 kg particle. Correct equation

M1A1

$$F = 58.1 - 19.5 = 38.6$$

Correct F

A1

$$R = 13.98 = 127.4$$

Correct R

B1

$$38.6 = \mu \times 127.4$$

Use of $F = \mu R$

dM1

$$\mu = \frac{38.6}{127.4} = 0.303$$

Correct coefficient of friction

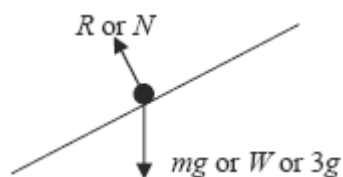
A1

6

[13]

Q20.

(a) (i)



Correct diagram with arrows and labels

B1

1

(ii) $3a = 3g \sin 30^\circ$

Two term equation of motion

M1

$$a = g \sin 30^\circ = 4.9 \text{ m s}^{-2}$$

AG Correct acceleration from correct working (Allow $a = g \sin 30^\circ$)

A1

2

(b) (i) $5 = \frac{1}{2} a \times 2^2$

Constant acceleration equation with $u = 0$

M1

$a = 2.5 \text{ m s}^{-2}$

*AG Correct answer from correct working.
(Use of $v = 5$ must be justified)*

A1

2

(ii) $3 \times 2.5 = 3g \sin 30^\circ - F$

Three term equation of motion

M1

Correct equation

A1

$F = 3g \sin 30^\circ - 7.5$
 $= 7.20 \text{ N (to 3 sf)}$

*Correct F
Accept 7.2 N*

A1

3

(iii) $R = 3g \cos 30^\circ (= 25.46)$

Resolving perpendicular to the slope to find R

M1

Correct R

A1

$7.2 = \mu \times 3g \cos 30^\circ$

Use of $F = \mu R$

M1

Correct expression

A1F

$$\mu = \frac{7.2}{3g \cos 30^\circ} = 0.283$$

Correct μ

Accept 0.282

(Follow through from incorrect F from above, but not an incorrect R)

A1F

5

- (iv) Reduce a , as the air resistance would reduce the magnitude of the resultant force or because the air resistance increases as the velocity increases towards its terminal value

Reduces

B1

Explanation

Second B1 dependent on the first B1 mark

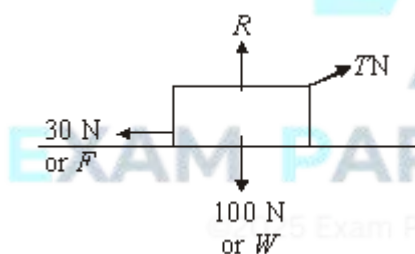
B1

2

[15]

Q21.

(a)



B1

1

(b) $F = \mu R, \quad 30 = 0.5 R$

M1

$R = 60 \text{ N}$

A1

2

- (c) Constant speed $\Rightarrow$ no acceleration / forces

M1

balance $\therefore 30\text{N} = T_x$

A1

2

(d) $T_y + R = 100$

M1

$T_v = 40$

A1

$T = \sqrt{(30^2 + 40^2)}$

Use of a vertical component of T

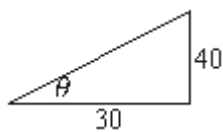
M1

$T = 50$

A1

4

(e)



$\tan \theta = \frac{40}{30}$

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M1

A1ft

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$\theta = 53.1^\circ$ Accept 53°

CAO

A1

3

[12]

Q22.

(a) (i) $R = 80\cos 25^\circ$

component attempted

M1

correct component

A1

$$R = 72.5\text{N}$$

CAO

A1

3

(ii) $F = 0.32 \times 72.5$
condone inequality

M1

$$F = 23.2\text{N}$$

CAO

A1

2

(iii) $T + F = 80\cos 65^\circ$
3 forces direction correct, component

M2

attempted

$$T = 10.6\text{N}$$

ft friction

A1

A1ft

4

(iv) $T = F + 80\cos 65^\circ$
3 forces, direction correct, component

M1

attempted

A1

$$T = 57.0\text{N} \text{ (57N)}$$

ft friction

A1ft

3

(iv) $\text{Mass} = \frac{80}{g} = (8.16\text{kg})$

B1

1

(b) $80 \cos 65^\circ - F = \text{mass} \times \text{acceleration}$

3 terms, component attempted

M1

$$10.6 = \frac{80}{g} \times \text{acc}$$

all correct

A1

$$\text{acc} = 1.30 \text{ ms}^{-2} (1.3 \text{ ms}^{-2})$$

CAO

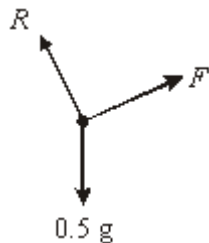
A1

3

[16]

Q23.

(a)



Accept W or mg (or 4.9) for weight
Arrows and labels needed
Can replace W with 2 correct components

B1

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(b) $F = 0.5g \sin 16^\circ$

component of weight attempted

M1

All correct including signs

A1

$$F = 1.35 \text{ N}$$

CAO

A1

3

(c) $R = 0.5g \cos 16^\circ$

component of weight attempted

M1

$$R = 4.71\text{N}$$

CAO

(b) and (c): For resolving horizontal/vertical M1
for each equation, A1 one equation correct

F A1 R A1

A1

2

(d) $F \leq \mu R$
 $1.35 \leq \mu \times 4.71$

Use of candidate's R, accept equals sign

M1

$$\mu \geq 0.287$$

CAO inequality required

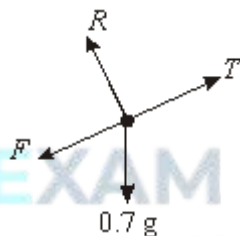
A1

2

[8]

Q24.

(a)



Accept W or mg (or 6.86) for weight

Arrows and labels needed

(can replace W with 2 correct components)

B1

1

(b) $R = 0.7g \cos 22^\circ$

component of weight attempted

M1

$$R = 6.36\text{ N}$$

all correct, including signs

A1

CAO

A1

3

$$F = 0.25 \times 6.36$$

(c) $F = 1.59 \text{ N}$

CAO

M1 A1

2

(d) $5.6 - 0.7g \sin 22^\circ$

$$a = 2.06 \text{ ms}^{-2}$$

4 terms with weight component attempted

M1

A marks -1 each error, accept $\pm 0.7a$

A2

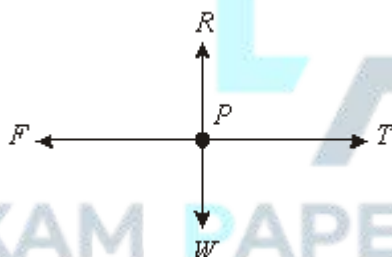
FT one error, accept $\pm$

A1F

[10]

Q25.

(a) (i)



Accept mg , $0.4g$ or 3.92 for weight

Arrows and labels needed

B1

1

(ii) $F = 0.5 \times (0.4 \times 9.8)$

Need to see 0.4×9.8 or 3.92 used

M1

$$F = 1.96 \text{ N}$$

A1

2

(b) $T - 1.96 = 1.4a$

$$0.3g - T = 0.3a$$

M1A1

Consistent reversal of signs in both equations
4 marks; reversal of signs in one equation,
M1 A1 M1 A0

M1A1

$$a = 1.4\text{ms}^{-2}$$

Sign change needs justification (whole string:
equation, $0.3g - 1.96 = 0.7a$ M1A1 $a = 1.4$ A1)
max 3/5

A1

5

(c) $v = 14 \times 3$
 $v = 4.2\text{ms}^{-2}$

Full method

M1

CAO

A1

2

- (d) P : Friction will cause
speed to decrease

M1

Accept decelerate or comes to rest

A1

- Q : Gravity will cause
speed to increase

M1

Accept accelerate

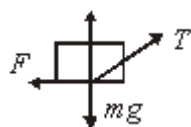
A1

4

[14]

Q26.

(a)



Correct diagram

B1

1

(b) $40 \cos 30^\circ - F = 25 \times 0.1$
 $F = 40 \cos 30^\circ - 2.5 = 32.1 \text{ N}$

Three term equation of motion

M1

Correct equation

A1

AG; correct force from correct working

A1

3

(c) $R + 40 \sin 30^\circ = 25 \times 9.8$
 $R = 225 \text{ N}$

Resolving vertically

M1

Correct equation

A1

AG; correct force from correct working

A1

3

(d) $32.1 = 225\mu$

use of $F = \mu R$

M1

$$\mu = \frac{32.1}{225} = 0.143$$

Correct μ

A1

2

[9]

Q27.

(a) $F = 0.8 \times 8 \times 9.8 = 62.72 \text{ N}$

Use of $F = \mu R$ with $R = 8 \times 9.8$

M1

Correct F

A1

2

(b) $7g - T = 7a$

Three term equation of motion for one particle

M1

$$T - 62.72 = 8a$$

Correct equation

A1

$$7g - 62.72 = 15a$$

Three term equation of motion for other particle

M1

$$a = \frac{7g - 62.72}{15} = 0.392$$

Correct equation

A1

AG; correct a from correct working

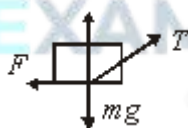
A1

5

[7]

Q28.

(a)



Correct diagram

B1

1

(b) $40 \cos 30^\circ - F = 25 \times 0.1$

$$F = 40 \cos 30^\circ - 2.5 = 32.1 \text{ N}$$

Three term equation of motion

M1

Correct equation

A1

AG; correct force from correct working

A1

3

(c) $R + 40 \sin 30^\circ = 25 \times 9.8$
 $R = 225 \text{ N}$

Resolving vertically

M1

Correct equation

A1

AG; correct force from correct working

A1

3

(d) $32.1 = 225\mu$

use of $F = \mu R$

M1

$$\mu = \frac{32.1}{225} = 0.143$$

Correct μ

A1

2

(e) Friction will decrease as normal reaction

Decrease in friction

B1

decreases

Normal reaction decreases

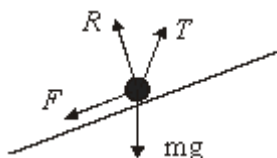
B1

2

[11]

Q29.

(a)



B1: Correct force diagram

B1

1

(b) $R + 20 \sin 30^\circ = 6g \cos 10^\circ$

M1: Resolving perpendicular to the slope with 3 terms

A1: Correct equation

M1A1

$$R = 6g \cos 10^\circ - 20 \sin 30^\circ$$

dM1: Solving for R

dM1

$$R = 47.9 \text{ N (to 3 sf)} \quad \text{AG}$$

A1: Correct R from correct working

A1

4

(c) $F = \mu R$

M1: Use of $F = \mu R$

M1

$$6 \times 0.4 = 20 \cos 30^\circ - 6g \sin 10^\circ - \mu R$$

M1: Resolving parallel to slope to get 4 term equation of motion

A1: Correct equation

M1A1

$$\mu R = 4.710$$

A1: Correct $F / \mu R$

A1

$$\mu = \frac{4.710}{47.91} = 0.0983$$

dM1: Solving for μ

dM1

A1: AWR 0.098

A1

6

[11]

Q30.

(a) $R = 2 \times 9.8, \quad F = \frac{1}{7} \times 2 \times 9.8 = 2.8 \text{ N}$

CAO M1: full method with 2×9.8 ,
accept inequality

M1A1
2

(b) (i) $P = 1, \quad F = 1\text{N}$

B1
1

(ii) $P = 5, \quad F = 2.8$
Used

B1

$$5 - 2.8 = 2a$$

M1: 3 terms, with forces subtracted

M1A1

$$a = 1.1 \text{ ms}^{-2}$$

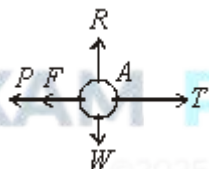
ft one slip

A1F
4

[7]

Q31.

(a) (i)



For any 3 forces correct

B1

For the rest correct, and no extras
(accept F in either direction, but not μR)

B1
2

(ii) $R = 0.2 \times g$

$$F = 0.5 \times R$$

For equations for both R and F

M1

$$F = 0.98\text{N}$$

$$AG$$

A1

2

(iii) $T = 0.3 \times 9.8 = 2.94\text{N}$

M1A1

2

(iv) $P + F = T$

M1

least $P = 1.96\text{N}$
ft T provided $P > 0$

A1F

2

(b) (i) $2.94 - T = 0.3a$

M1A1

$T - 0.98 = 0.2a$
M1 either equation
(whole string method: 'correct' equation, M1 A1,
answer A1; max 3 / 5)

A1

$1.96 = 0.5a$

m1

$a = 3.92\text{ms}^{-2}$
ft one error

A1F

5

(ii) $0.1 = \frac{1}{2} \times 3.92 \times t^2$
Use of candidate's acceleration, if 9.8

M1

$t = 0.226$ (0.22588)
ft candidate's acceleration (not 9.8) if
 $a = 1.96$ used, $t = 0.319$

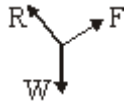
A1F

2

[15]

Q32.

(a)



Three forces labelled & with arrows, W , mg or 35 g vertical (or 2 components of W), R and F perpendicular (ignore pairs of components of existing forces)

B1

1

(b) $R = 35g \times \cos 25^\circ$

Component attempted & g present for M1

M1A1

$R = 311\text{ N}$

($R = 310.86$) accept AWRT 311

A1F

3

(c) $F = 35g \times \sin 25^\circ$

Component attempted & g present for M1 & acceleration zero

M1A1

($= 144.96$)

$F = \mu R$, $144.96 = \mu \times 310.86$

Use of friction law with candidate's values, must have tried to find F

M1

$\mu = 0.466$

ft R and F , provided $\mu > 0$

M1A0 if $F < \mu R$ used

SC accept use of $\mu = \tan \theta$

A1F

4

[8]

Q33.

(a) $v^2 = u^2 + 2as$

Use of full method

M1

$$0^2 = 3.5^2 + 2 \times a \times 2.5$$

Correct subs

A1

$$a = -2.45 \text{ mag } 2.45$$

Magnitude required

A1

3

(b) (i) Friction : $0.2 \times 2.45 = 0.49\text{N}$

M1: use of $F = ma$ with ± 2.45

A1: Magnitude required

M1A1

2

(ii) $R = 0.2 \times g$

$$F = \mu R \quad 0.49 = \mu \times 0.2g$$

Use of $F = \mu R$ with $R = mg$ substituted

M1

$$\mu = 0.25$$

f.t. (i) provided μ positive

Use of $F < \mu R$, M1 A0

A1F

2

[7]

Q34.

(a) (i) $B: T = mg = 0.1 \times 9.8 = 0.98\text{N}$

cao

B1

1

(ii) $A: F = T (= 0.98) \quad R = 0.5 \times 9.8$

B1

$$\mu \times 0.5 \times 9.8 = 0.98$$

Condone inequality

M1

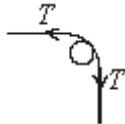
$$\mu = 0.2$$

cao

A1

3

(iii)

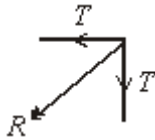


Must label T or 0.98 N

B1

1

(iv)



$$R = \sqrt{T^2 + T^2} \text{ or } 2T \cos 45^\circ$$

M1

$$R = 1.39\text{N}$$

A1

2

(b) (i) $A: T - 0.98 = 0.5a$

Either equation for M1

M1A1

$$B: 2 \times 0.1 \times 9.8 - T = 0.2a$$

A1

$$0.98 = 0.7a$$

m1

$$a = 1.4$$

cao

A1
5

- (ii) $T = 0.98 + 0.5 \times 1.4$, $T = 1.68$
ft accuracy error in equation

A1F
1

(iii) $s = ut + \frac{1}{2} at^2$

$$0.7 = 0 + \frac{1}{2} \times 1.4t^2$$

M1

$$t = 1$$

A1
2

[15]

Q35.

- (a) (i) $T = mg$
Accept $g = 9.8$ (or 10) used throughout (a)

B1
1

(ii) $F = T (= mg)$

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$$R = 2mg$$

B1

$$F \leq \mu R, \quad mg \leq 2mg\mu, \quad \mu \geq \frac{1}{2}$$

*Inequality needed, or convincing verbal explanation,
 ft if μ numeric and positive*

B1F
3

(b) $T = 2ma$

M1 either equation

M1A1

$$mg - T = ma$$

Accept $g = 9.8$ or 10 for either A1

A1

$$2mg - 2T = T$$

Full method for T

m1

$$T = \frac{2mg}{3}$$

ft one slip

If friction included, can either earn M1A1 for Q ,
A0 for P , m1A0.

SC Whole string method can earn m1 for $mg = 3ma$
and A1F for result for T if M1 already earned

A1F

5

(accept: $T = 6.53m$ if 9.8 used, $T = 6.67m$ if 10 used,
 $T = 6.54m$ if 9.8 used and a found first)

[9]